

Softmax from Fast Mixing

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Abstract

Softmax is used as a choice law almost everywhere, usually without a mechanism that produces it. We give one. When N channels compete under a shared hidden state that mixes faster than the channels fire, the winning probability is softmax in the stationary mean intensities at leading order, and the first correction is a Green–Kubo term built from integrated rate autocovariances. One pair of objects, the mean intensities and the correlation matrix K , then predicts every blocked-subset counterfactual by restricting sums, so the whole deletion ensemble follows from one calculation. Fluctuations common to all channels cancel exactly, so departures from softmax measure differential structure alone, and environmental modes of rank r give K of rank exactly r . We verify the expansion on finite chains, then on Brownian narrow escape with reaction windows, where the leading and corrected errors decay with measured orders 0.93 and 1.93 against an exactly discretized generator.

1 Introduction

A choice among competing options is often modelled by softmax. The rule assigns option i a probability proportional to e^{θ_i} , and it appears in discrete choice, reinforcement learning, statistical physics, and the output layer of essentially every classifier. Its usual justification is convenience. Softmax is smooth, normalized, and differentiable, and in the Gumbel random-utility model it is exact.

Underneath many applications, though, there is real kinetics. A chemical reaction fires when one of several channels crosses a barrier first. A diffusing particle escapes through whichever window it encounters first. A kinetic Monte Carlo (KMC) step selects among escape channels by their rates. In these systems the choice is a race, and the race has a mechanism. It is worth asking what that mechanism implies about the choice law, rather than assuming softmax and moving on.

This paper answers that for one specific and common situation. The channels share a hidden state, and that state mixes fast relative to the rate at which channels fire. Catalysis supplies the picture. Adsorbate configurations rearrange quickly while the reaction event is rare, so every channel sees an environment that has effectively averaged out. Under that separation we show two things.

The leading-order result is that softmax is exact in the limit, with $\theta_i = \log \bar{\lambda}_i$ the log of the stationary mean intensity. Fast mixing homogenizes the environment, each channel is left with

*Code and machine verification for every claim in this paper: <https://github.com/microprediction/kinetics> (experiments 7 and 9). Web companion: <https://kinetics.microprediction.org>.

its average rate, and the race among constant rates is the exponential race, whose choice law is Luce.

The first correction is a Green–Kubo term. It is built from $K_{jk} = \int_0^\infty \text{Cov}_\pi(\lambda_j(Y_0), \lambda_k(Y_t)) dt$, the time-integrated covariance of the firing intensities under the stationary law. This is a dynamical object rather than a static one. Two channels whose rates are correlated at equal times contribute nothing unless the correlation persists over the timescale on which the environment decorrelates.

The correction is what breaks independence of irrelevant alternatives (IIA). Softmax redistributes a removed channel’s probability in proportion to the survivors’ probabilities, which is the IIA prediction and is usually wrong in physical systems. The Green–Kubo term supplies the deviation, and it does so for every blocked subset at once.

That last point is the practical one. The pair $(\bar{\lambda}, K)$ is computed once. Any counterfactual, remove channel 3, block channels $\{2, 5, 7\}$, is answered by restricting the sums in the correction to the surviving set. No new solve is required per counterfactual. This is the statistical counterpart of a theme running through our related work, that one global computation can contain an entire leave-one-out ensemble.

Two structural facts sharpen the result. Fluctuations shared by all channels, $\lambda_i(y) = a_i c(y)$, cancel from the correction identically. We show this is not merely an asymptotic statement. A common fluctuation is a common time change, so Luce holds exactly at every degree of separation, which is the proportional-hazards normal form. Separately, if the intensity deviations are driven by r environmental modes then K has rank exactly r , so the correction inherits any low-dimensional structure the physics has.

Section 2 sets up the shared-state race and the killed resolvent. Section 3 states and proves the expansion. Section 4 collects the three structural properties. Section 5 verifies everything on finite chains, where each quantity is an exact linear solve. Section 6 moves to continuous physics, a reflected Brownian motion in a disk with reaction windows, and reports the measured convergence orders together with two systematic errors that a layered numerical design exposed.

2 The shared-state race

Let Y_t be an ergodic Markov process on a state space with generator L/ε and invariant law π . The parameter ε is the ratio of the environment’s mixing time to the timescale on which channels fire. Small ε means fast mixing.

Each channel $i \in \{1, \dots, N\}$ fires at intensity $\lambda_i(Y_t) \geq 0$, conditionally Poisson given the path of Y . The first channel to fire wins. For an available set $A \subseteq \{1, \dots, N\}$ write $\Lambda_A(y) = \sum_{j \in A} \lambda_j(y)$ for the total hazard, and write $\bar{\lambda}_i = \int \lambda_i d\pi$ for the stationary mean intensity.

Let $u_i^A(y)$ be the probability that channel i wins, started from $Y_0 = y$. Conditioning on the first instant and using the conditional Poisson structure gives

$$\left(\frac{L}{\varepsilon} - \Lambda_A\right) u_i^A = -\lambda_i, \quad u_i^A = \left(\Lambda_A - \frac{L}{\varepsilon}\right)^{-1} \lambda_i. \quad (1)$$

This is a killed resolvent. The generator propagates the environment and the multiplication operator Λ_A kills paths at the total firing rate. On a finite state space it is a linear system, which is what makes the verification in Section 5 exact.

The observable of interest is the win probability from a stationary start,

$$p_i^A = \int u_i^A d\pi. \quad (2)$$

Summing (1) over $i \in A$ gives $\sum_{i \in A} u_i^A = 1$, so the p_i^A are a probability vector on A for every ε .

3 Homogenization and the Green–Kubo correction

Write $\tilde{\lambda}_i = \lambda_i - \bar{\lambda}_i$ for the fluctuation of the i th intensity, and Π for the projection $\Pi f = (\int f d\pi) \mathbf{1}$ onto constants. The deviation operator

$$D = (\Pi - L)^{-1}(I - \Pi) \quad (3)$$

inverts L on the mean-zero subspace, and $Dg = \int_0^\infty e^{tL} g dt$ for mean-zero g . Note the sign it carries. Differentiating that integral gives $LDg = -g$, so D is minus the pseudo-inverse of L , and the sign propagates into the proof below. The Green–Kubo matrix is

$$K_{jk} = \int_0^\infty \text{Cov}_\pi(\lambda_j(Y_0), \lambda_k(Y_t)) dt = \int \tilde{\lambda}_j (D\tilde{\lambda}_k) d\pi. \quad (4)$$

The second expression is what one computes. It replaces a time integral by a single linear solve. Note that K need not be symmetric. It is symmetric when the environment is reversible, and the verification in Section 5 deliberately uses a non-reversible chain so that the index placement in the theorem is actually tested.

Theorem 1 (Homogenized choice law and its first correction). *Let $\bar{\Lambda}_A = \sum_{j \in A} \bar{\lambda}_j$ and suppose $\bar{\Lambda}_A > 0$. Then as $\varepsilon \rightarrow 0$,*

$$p_i^A = \frac{\bar{\lambda}_i}{\bar{\Lambda}_A} - \frac{\varepsilon}{\bar{\Lambda}_A} \left[\sum_{j \in A} K_{ji} - \frac{\bar{\lambda}_i}{\bar{\Lambda}_A} \sum_{j, k \in A} K_{jk} \right] + O(\varepsilon^2). \quad (5)$$

In particular $p_i^A \rightarrow \bar{\lambda}_i / \bar{\Lambda}_A$, the softmax law in the log mean intensities $\theta_i = \log \bar{\lambda}_i$.

Proof. Multiply (1) by ε to get $(L - \varepsilon \Lambda_A) u_i^A = -\varepsilon \lambda_i$, expand $u_i^A = u_i^{(0)} + \varepsilon u_i^{(1)} + \varepsilon^2 u_i^{(2)} + O(\varepsilon^3)$, and collect powers of ε .

At order ε^0 , $Lu_i^{(0)} = 0$, so $u_i^{(0)}$ is a constant c_i by ergodicity.

At order ε^1 the equation reads $Lu_i^{(1)} = \Lambda_A c_i - \lambda_i$. A solution exists only if the right side is orthogonal to π , which forces $\bar{\Lambda}_A c_i = \bar{\lambda}_i$ and gives the leading term. The right side is then $c_i \tilde{\Lambda}_A - \tilde{\lambda}_i$ with $\tilde{\Lambda}_A = \sum_{j \in A} \tilde{\lambda}_j$, and inverting with $LD = -I$ on mean-zero functions gives

$$u_i^{(1)} = -D(c_i \tilde{\Lambda}_A - \tilde{\lambda}_i) + m_i,$$

where the constant $m_i = \int u_i^{(1)} d\pi$ is fixed at the next order.

At order ε^2 the equation is $Lu_i^{(2)} = \Lambda_A u_i^{(1)}$, whose solvability condition is $\int \Lambda_A u_i^{(1)} d\pi = 0$. The constant part of Λ_A contributes $\bar{\Lambda}_A m_i$, the constant m_i integrates against $\tilde{\Lambda}_A$ to zero, and what remains is

$$0 = \bar{\Lambda}_A m_i - \int \tilde{\Lambda}_A D(c_i \tilde{\Lambda}_A - \tilde{\lambda}_i) d\pi.$$

Using (4), $\int \tilde{\Lambda}_A D\tilde{\lambda}_i d\pi = \sum_{j \in A} K_{ji}$ and $\int \tilde{\Lambda}_A D\tilde{\Lambda}_A d\pi = \sum_{j,k \in A} K_{jk}$, so

$$m_i = -\frac{1}{\bar{\Lambda}_A} \left[\sum_{j \in A} K_{ji} - \frac{\bar{\lambda}_i}{\bar{\Lambda}_A} \sum_{j,k \in A} K_{jk} \right].$$

Finally $p_i^A = c_i + \varepsilon m_i + O(\varepsilon^2)$ by (2), which is (5). $\square$

Remark 1. The two terms in the bracket have distinct roles. The first, $\sum_{j \in A} K_{ji}$, measures how channel i 's rate correlates dynamically with the total hazard it competes against. The second subtracts the part of that effect which is common to all channels, and is what keeps $\sum_{i \in A} p_i^A = 1$ at each order.

4 Three structural properties

One computation answers every counterfactual. The pair $(\bar{\lambda}, K)$ does not depend on A . Every blocked-set counterfactual is obtained from (5) by restricting the sums to the surviving set. For a system with N channels this replaces 2^N separate resolvent solves with one deviation solve and $O(N^2)$ stored numbers.

Common-mode fluctuations cancel. Suppose $\lambda_i(y) = a_i c(y)$, so that every channel is modulated by the same environmental factor. Then $\lambda_i = a_i \tilde{c}$ and $K_{jk} = a_j a_k \gamma$ with $\gamma = \int \tilde{c}(D\tilde{c}) d\pi$. Substituting into the bracket in (5) gives $a_i \gamma \sum_{j \in A} a_j - (a_i / \sum_{j \in A} a_j) \gamma (\sum_{j \in A} a_j)^2 = 0$. The correction vanishes identically.

Proposition 1 (Proportional hazards give Luce exactly). *If $\lambda_i(y) = a_i c(y)$ with $c > 0$ then $p_i^A = a_i / \sum_{j \in A} a_j$ exactly, for every ε and every A , not merely to the order computed above.*

Proof. Let $\tau_c(t) = \int_0^t c(Y_s) ds$, a strictly increasing time change. In the clock τ_c every channel fires at the constant rate a_i , because the cumulative hazard of channel i up to real time t is $a_i \tau_c(t)$. The race is then a race among constant exponential rates, whose winner is i with probability $a_i / \sum_{j \in A} a_j$, independently of the clock. Since the identity of the winner is invariant under a common monotone time change, the same holds in real time. $\square$

This is a stronger statement than the vanishing of the correction, and it is practically useful in the opposite direction. It says a measured departure from softmax cannot be produced by any amount of shared environmental fluctuation. Whatever deviation is observed is evidence of differential structure among the channels.

Low-rank environments give low-rank corrections. Suppose the intensity deviations are driven by r environmental modes, $\lambda_i - \bar{\lambda}_i = b_i^\top z(y)$ with z taking values in $\mathbb{R}^r$ and $\int z d\pi = 0$. Then $K = B\Gamma B^\top$ with B the $N \times r$ matrix of loadings and $\Gamma_{ab} = \int z_a(Dz_b) d\pi$, so K has rank at most r . The compressed correlated race that appears elsewhere in this program therefore has a physical provenance rather than being a modelling convenience.

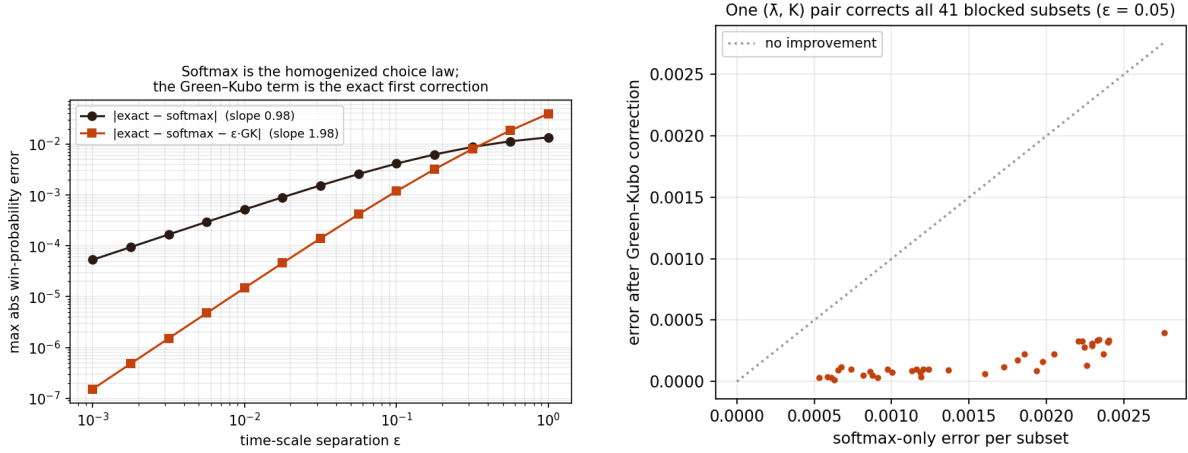


Figure 1: Finite-chain verification (experiment 7). Left, error against ε for the leading softmax law and for the Green–Kubo corrected law, with measured orders 0.985 and 1.985. Right, the same $(\bar{\lambda}, K)$ pair applied to 41 random availability sets at $\varepsilon = 0.05$, with no per-subset computation.

5 Verification on finite chains

On a finite state space every object above is an exact linear solve, so the theorem can be tested without simulation error. Experiment 7 in the repository runs a six-state chain with ten channels.

The measured convergence order of the leading term is 0.985, against the theoretical 1. After adding the Green–Kubo correction the measured order is 1.985, against the theoretical 2. The gain of exactly one order is the signature the expansion predicts.

The subset-uniformity claim is tested by fixing one $(\bar{\lambda}, K)$ pair and predicting 41 random availability sets at $\varepsilon = 0.05$. The maximum error over all subsets falls from 2.8×10^{-3} under softmax to 4.0×10^{-4} with the correction, with no per-subset computation.

Common-mode cancellation is confirmed at 2.6×10^{-16} for the Green–Kubo coefficient, and the exact win probabilities agree with Luce to 1.1×10^{-16} at $\varepsilon = 0.3$, which is far outside the asymptotic regime and is the content of Proposition 1. Rank-two environmental loadings give K with singular values $[0.022, 0.002, 0, 0]$, so the rank is exactly two.

A Gillespie simulation of the same system, event-driven and including the hidden state, agrees with the resolvent solution to 1.9×10^{-3} across 40,000 trajectories at $\varepsilon = 0.3$, which is its own sampling noise.

An independent reimplement, written from the statement of Theorem 1 alone and run on a deliberately non-reversible chain where K is asymmetric by 4.0×10^{-3} , reproduces the order gain (0.934 to 1.935), matches K against brute-force time integration of the covariance to 7.7×10^{-10} , and reproduces both structural properties. The non-reversible test matters because it is what distinguishes K_{ji} from K_{ij} in (5).

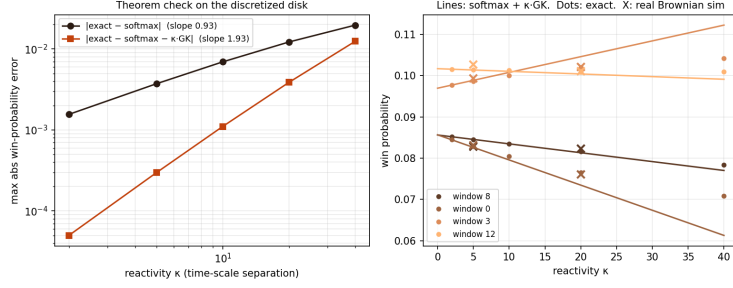


Figure 2: The theorem on continuous physics (experiment 9). Reflected Brownian motion in the unit disk with boundary reaction windows, error against κ for the leading and corrected laws, measured orders 0.93 and 1.93 against an exactly discretized Neumann generator.

6 Narrow escape

Finite chains verify the algebra. To test the theorem as a statement about physics we take the hidden state to be reflected Brownian motion in the unit disk and the channels to be reaction windows on the boundary, bands in which channel i fires at rate κ . The first window to fire wins. Here κ plays the role of ε , since a slow reaction rate means the particle mixes many times before firing.

Experiment 9 separates three layers so that no error source can hide in another. Layer A is a real Brownian simulation with independent per-step firing. Layer B is an exact killed-resolvent solve on a finite-volume Neumann Laplacian over a polar grid, with the uniform invariant law, reversibility, and conservation asserted numerically rather than assumed. Layer C is the theory, band-area softmax plus the Green-Kubo term from one deviation solve, using the same code that passed the chain tests.

Comparing layers B and C tests the theorem on a continuous-physics generator. The softmax error decays with measured order 0.93 and the corrected error with order 1.93, again a clean gain of one order. At $\kappa = 2$ the correction takes the maximum error from 1.6×10^{-3} to 5.0×10^{-5} .

Comparing layers A and B tests the discretization. The maximum discrepancy is 2.2×10^{-3} at $\kappa = 5$ with 50,000 trajectories against a sampling noise of about 10^{-3} , and falls to 9.6×10^{-4} against a noise floor of 7×10^{-4} in a 200,000-trajectory check.

Two systematic errors the layering caught. Both would have corrupted a direct comparison of simulation against theory, and neither is visible without the middle layer.

The first is band quantization, worth 5×10^{-2} . Sharp cell indicators mis-size any window narrower than a grid cell, by up to a factor of three. Fractional area-exact cell coverage fixes it, after which band areas match their analytic values to 10^{-14} .

The second is overlap semantics, worth 1.4×10^{-2} . The window geometry contains three overlapping pairs. The original simulation awarded an overlap to the lower-indexed window, while the model stacks intensities as $\lambda = \kappa \sum_i g_i$, which is the physically correct convention for independent receptors. The error's structure identified it, being -0.014 on one window and $+0.006$ on its overlap partner, and its flatness under both time-step and grid refinement confirmed that it was not a discretization artifact.

7 Discussion

The result explains why softmax is a reasonable default in systems with fast shared dynamics, and it quantifies exactly when that default fails. The failure is governed by integrated dynamical correlations, not by static ones, and it is insensitive to any fluctuation shared across channels.

Three limitations bound the claim. The expansion is in the mixing-to-firing timescale ratio and says nothing when channels fire faster than the environment relaxes. The correction is first order, so systems at moderate separation retain an $O(\varepsilon^2)$ residual, visible in the finite- κ end of Section 6. The results are proved for finite chains and verified on a reversible diffusion, and a formal treatment of reversible diffusions with a spectral gap is left open.

Two extensions look tractable. The disk’s boundary operator diagonalizes in Fourier modes, which may render the leading geometric correction analytic rather than numerical. Separately, the Green–Kubo kernel and the empirical substitution kernel of the companion work describe the same non-IIA structure from theory and from data, and identifying them would connect a computed correction to an estimated one.